

IE4497/EE6497 Pattern Recognition and Deep Learning

Homework 1

To be submitted by 11 Sep. 2026, 23:59

1. (10 marks) Provide *short answers* to each of the following.

- (a) Suppose $\log p(x) = ax + b$ for $x \geq 0$, where a, b are constants with $a < 0$. What famous pdf is $p(x)$?
- (b) Conditioned on $z \sim \text{Cat}(\cdot | 0.5, 0.4, 0.1)$, x has exponential distribution $\text{Exp}(x | \lambda_z)$, where $\lambda_1 = 7.8$, $\lambda_2 = 0.6$ and $\lambda_3 = 4.5$. Is $p(x)$ a mixture model? Write down the pdf of x in terms of $\text{Exp}(\cdot | \cdot)$.
- (c) The R^2 value is defined by

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2},$$

where y_i is the i -th training response, $\hat{y}_i$ its predicted response, and $\bar{y} = \frac{1}{n} \sum_i y_i$. Can $R^2 < 0$? Explain how this can happen.

2. (25 marks) A hidden Markov model (HMM) has latent state space $\mathcal{Z} = \{s_1, s_2\}$ and emission symbols $\{o_1, o_2\}$. The initial state probabilities are $\boldsymbol{\pi}(s_1) = \boldsymbol{\pi}(s_2) = 1/2$. The state transition matrix is

$$\mathbf{T} = \begin{bmatrix} a & 1-a \\ 1-b & b \end{bmatrix},$$

and the emission probabilities are given by

$$\begin{aligned} p(o_1 | s_1) &= p, & p(o_2 | s_1) &= 1-p, \\ p(o_1 | s_2) &= q, & p(o_2 | s_2) &= 1-q. \end{aligned}$$

For a sequence of length L , let $\mathbf{z}[1 : L] = (\mathbf{z}[1], \dots, \mathbf{z}[L])$ denote the latent states and $\mathbf{x}[1 : L] = (\mathbf{x}[1], \dots, \mathbf{x}[L])$ denote the corresponding emissions.

In a HMM, only the emission sequence is observed. On the other hand, if the latent states are observed in addition to the emissions, this is a *standard Markov model with emissions*. In this model, an observed sequence of length L consists of the pair $(\mathbf{z}[1 : L], \mathbf{x}[1 : L])$.

- (a) Suppose observations over $L = 2$ time steps are made.

- (i) Consider the standard Markov model with emissions, where the emission sequence is $\mathbf{x}[1 : 2] = (\mathbf{x}[1] = o_2, \mathbf{x}[2] = o_1)$. Derive an expression for the joint probability $p(\mathbf{z}[1], \mathbf{z}[2], \mathbf{x}[1] = o_2, \mathbf{x}[2] = o_1)$ in terms of a, b, p , and q for each possible combination of $\mathbf{z}[1]$ and $\mathbf{z}[2]$.
- (ii) Consider the HMM in which only the emission sequence is observed. Derive an expression for the likelihood of the emission sequence $(\mathbf{x}[1] = o_2, \mathbf{x}[2] = o_1)$ in terms of a, b, p , and q .
- (b) Suppose p and q are known and $p \neq q$, but a and b are unknown. Derive the maximum likelihood estimate (MLE) for a and b in each of the following cases:
- (i) In the standard Markov model with emissions, the observed data consists of four independent sequences as follows:
- $(\mathbf{z}_1[1] = s_1, \mathbf{z}_1[2] = s_1, \mathbf{x}_1[1] = o_2, \mathbf{x}_1[2] = o_1);$
 - $(\mathbf{z}_2[1] = s_1, \mathbf{z}_2[2] = s_2, \mathbf{x}_2[1] = o_2, \mathbf{x}_2[2] = o_1);$
 - $(\mathbf{z}_3[1] = s_2, \mathbf{z}_3[2] = s_2, \mathbf{x}_3[1] = o_2, \mathbf{x}_3[2] = o_1);$
 - $(\mathbf{z}_4[1] = s_2, \mathbf{z}_4[2] = s_2, \mathbf{x}_4[1] = o_2, \mathbf{x}_4[2] = o_1).$
- Hint: You may use the expressions derived in part (a)(i).*
- (ii) In the HMM, a single emission sequence is observed as follows: $(\mathbf{x}_1[1] = o_2, \mathbf{x}_1[2] = o_1)$.
Hint: You may directly optimize the expression derived in part (a)(ii). Consider the cases where $p > q$ and $p < q$ separately.
- (c) In the HMM, suppose now that the observed data consists of 5 independent emission sequences, each of length $L = 20$. What algorithm would you use to find the MLE for a and b and why?

The following question is mandatory for students taking EE6497 and optional (bonus) for students taking IE4497.

- 3.** (15 marks) Consider a device whose time to failure is modeled as a random variable x taking values in the set $\{1, 2, \dots, r, r + 1\}$. The probability mass function (pmf) of x is

defined as follows:

$$p(x = k \mid \theta) = \theta^{k-1}(1 - \theta), \quad \text{for } k = 1, 2, \dots, r,$$

$$p(x = r + 1 \mid \theta) = \theta^r,$$

where r is a positive integer and $0 < \theta \leq 1$. Suppose the times to failure of n independent devices are observed, denoted as $x_1, x_2, \dots, x_n$. Let n_k represent the number of observed failure times x_i that take the value k , for $k = 1, 2, \dots, r + 1$.

- (a) Derive the log likelihood of the observed time to failure data $x_1, x_2, \dots, x_n$. Express your answer in terms of the quantities n_k , $k = 1, \dots, r + 1$, taking care that the probability of the event $x = r + 1$ has a different formula from the other events.
- (b) Show that the MLE of θ is given by

$$\theta_{\text{MLE}} = \frac{\sum_{k=1}^{r+1} kn_k - n}{\sum_{k=1}^{r+1} kn_k - n_{r+1}}.$$